LIMIT EVALUATION

$$\lim_{x\to4}\frac{2-\sqrt{x}}{4-x} = \lim_{x\to 4}\frac{(2-\sqrt{x})}{2^2-(\sqrt{x})^2} = \frac{1}{2+\sqrt{4}} = \frac{1}{4}$$

SOLVE THIS ONE AS CLASSWORK
AND COMMENT ANSWER IN CHAT.

$$\left[\frac{2-\sqrt{4}}{4-4} = \frac{2-2}{0} = \frac{0}{0} \text{ form}\right]$$

DIY

1) FACTORIZE → CANCELLATION

2) Multiplying by Conjugate by dividing

$(2-i) = 2+i \quad (2-\sqrt{x})(2+\sqrt{x})$

3) L'Hopital's Law/Rule

$$\lim_{x\to1} tan^{-1}(x) \overset{?}{=} \cancel{45^0} \text{ or } 45 \text{ degree} = \frac{\pi}{4} \text{ rad}$$

$$= \frac{\pi}{4}$$

SOLVE THIS ONE AS CLASSWORK
AND COMMENT ANSWER IN CHAT.

$\tan 45^0 = 1$

$\tan \frac{\pi}{4}^c$

$\tan\left(\frac{\pi}{4} \text{ rad}\right) = 1$

$\Rightarrow \tan^{-1}(1) = \pi/4$

$$\boxed{\lim_{x\to\infty} tan^{-1}(x) = \frac{\pi}{2}}$$

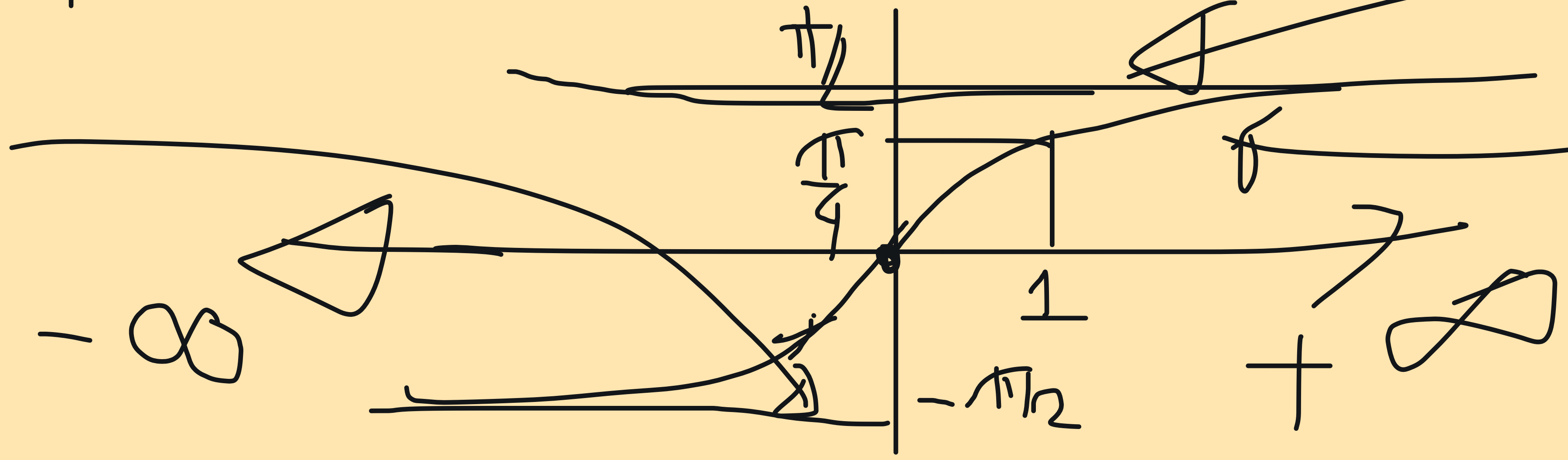
$\forall x, \quad y = \tan x$

$\tan^{-1} x \neq \left(\frac{1}{\tan}\right)$

$\tan^{-1} x \neq (\tan x)^{-1}$

$$\boxed{\lim_{x\to-\infty} tan^{-1}(x) = -\frac{\pi}{2}}$$

$\boxed{y = -\frac{\pi}{2}}$



$\boxed{y = \frac{\pi}{2}}$

$y = \tan^{-1}(x)$

LIMIT TO INFINITY

$$\lim_{x\to-\infty} \frac{2-\sqrt{x^2}}{4-x^2}$$

$2-\sqrt{(-\infty)^2}$

DIY $\Rightarrow 2-\sqrt{(-\infty)(-\infty)}$

$\Rightarrow 2-\infty$

$\Rightarrow -\infty$

$4-(\infty)^2 \Rightarrow 4-\infty = -\infty$

$\dfrac{-\infty}{-\infty} = \dfrac{\infty}{\infty}$ form

$$\lim_{x\to-\infty} \frac{2-\sqrt{x^2}}{4-x}$$

$= \lim_{x\to-\infty} \dfrac{2-(|x|)}{4-x}$

$\forall x \in \mathbb{R},$
$\Rightarrow x^2 \geq 0$
$\Rightarrow \sqrt{x^2} \geq 0$
$\Rightarrow |x| \geq 0$

$= \lim_{x\to-\infty} \dfrac{(2-(-x))}{(4-x)}$

$x=0$

$x<0, \quad x>0$

$-\infty \quad 0 \quad +\infty$

$|x| = \begin{cases} x, & x>0 \\ -x, & x<0 \end{cases}$

$\Downarrow$

$\lim_{x\to-\infty} \left[\dfrac{2+x}{4-x}\right] \Rightarrow$

DIY

TRY $\dfrac{x\left(\frac{2}{x}+1\right)}{x\left(\frac{4}{x}-1\right)}$

$\dfrac{1}{0} \neq \dfrac{0}{0}$ Indeterminate

$= \lim_{x\to-\infty} \dfrac{\frac{d}{dx}(2+x)}{\frac{d}{dx}(4-x)}$

$x \to -\infty$

$2+x \to 2-\infty \Rightarrow -\infty$

$4-x \to 4-(-\infty) \to \infty$

$\dfrac{-\infty}{\infty}$ form

$= \lim_{x\to-\infty} \dfrac{(0+1)}{(0-1)} \checkmark \quad \lim_{x\to-\infty}\left(\dfrac{1}{-1}\right) = -1$

INFINITE LIMIT

$\lim_{x\to\frac{\pi}{2}} tan\,(x) = DNE$

$\lim_{x\to\frac{\pi}{2}^+} (\tan x) = -\infty \notin \mathbb{R}$

$\lim_{x\to\frac{\pi}{2}^-} (\tan x) = +\infty \notin \mathbb{R}$

DNE

$\rightarrow$ DNE

$$\lim_{x\to 3} \frac{x^2-3x+3}{9-12x+6x^2-x^3}$$

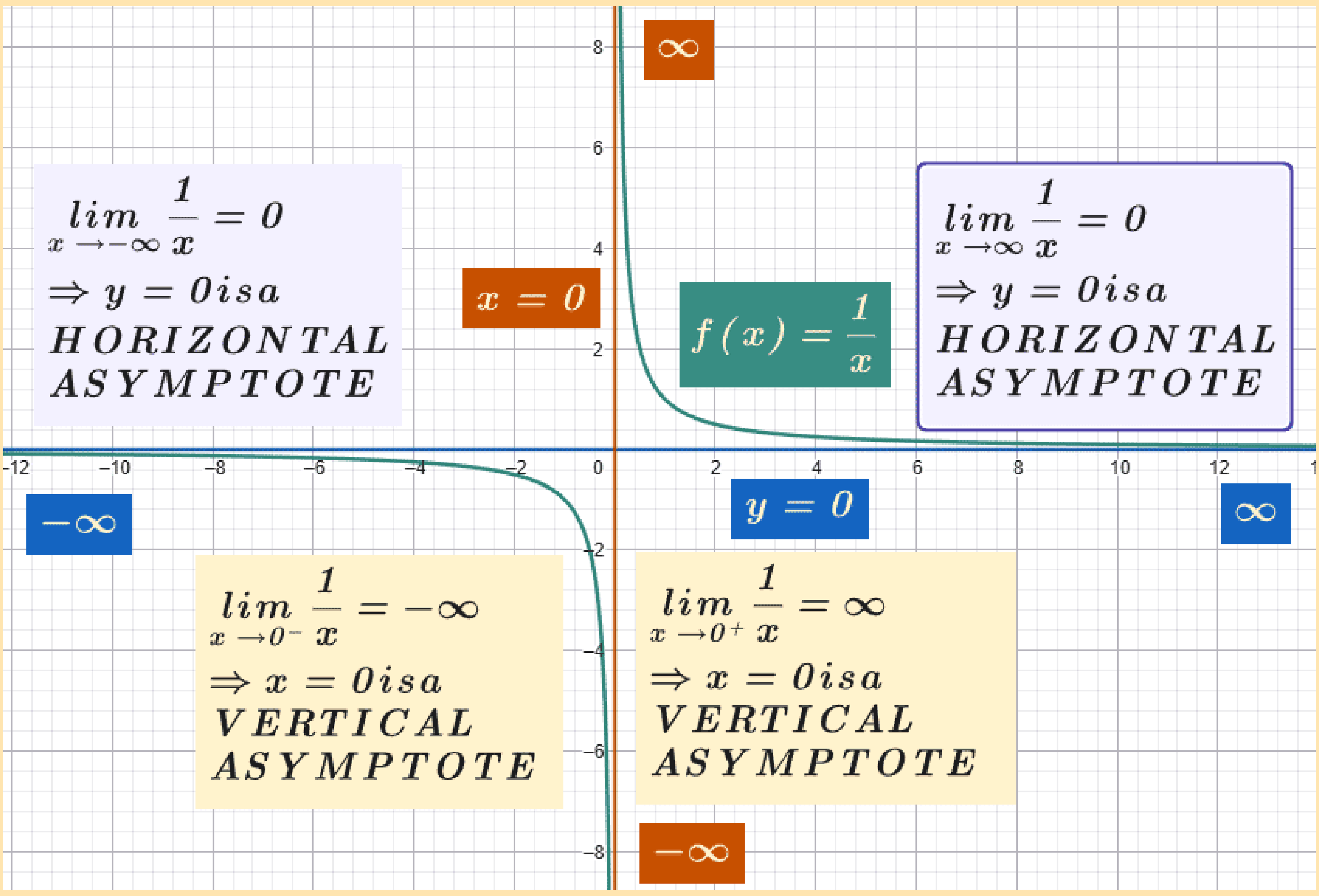
DIY

1. Direct CALCULATOR $x \to 3$
2. Long polynomial division
3. FACTORIZE 2 polynomial.
4. Let, $x = 3+h \Rightarrow h = x-3$

$\boxed{x\to 3 \Rightarrow h\to 0}$

$$\lim_{h\to 0} \frac{\left[(3+h)^2 - 3(3+h) + 3\right]}{9-12(3+h)+6(3+h)^2-(3+h)^2}$$

ASYMPTOTE

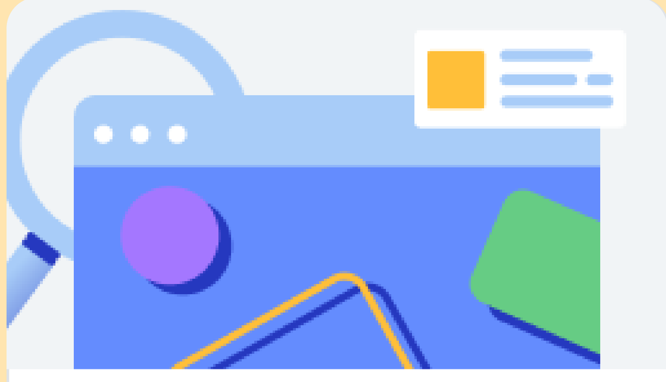


$$\lim_{x\to\frac{\pi}{4}} tan\,(2x)$$

DIY?

$$\lim_{x\to\infty} tan^{-1}\,(-x)$$

DIY?



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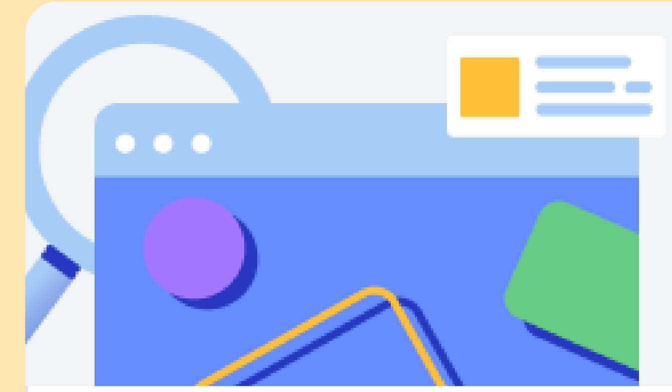
LIMIT: FORMAL DEFINATION

$$MATHEMATICALLANGUAGE$$

$$\forall \varepsilon > 0, L \in \Re, \exists \delta > 0 : 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

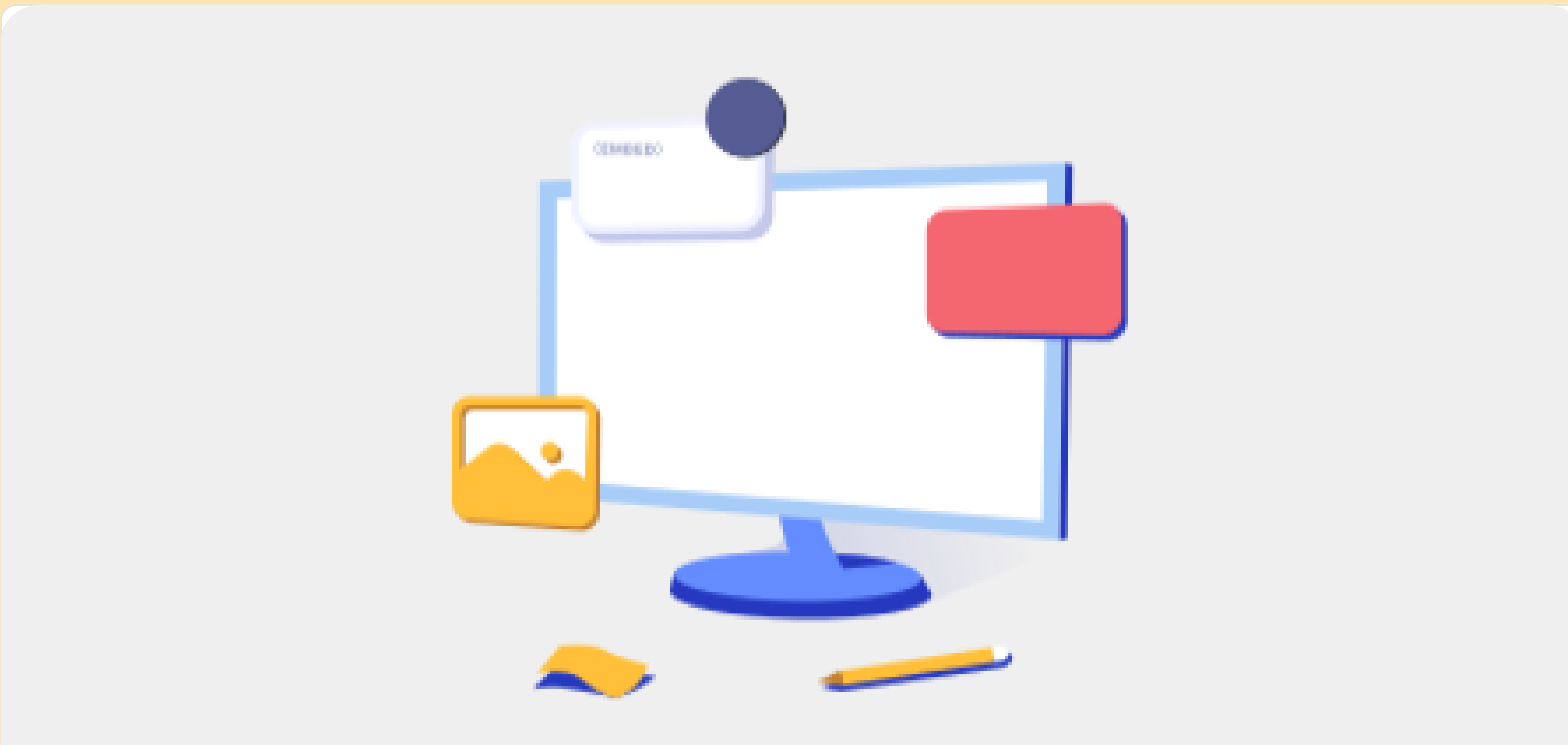
https://www.mathsisfun.com/calculus/limits-formal.html



Limits (Formal Definition)

Sometimes we can't work something out directly ... but we can see what it should be as we get closer and closer!

Math is Fun



Limits (Formal Definition)

Math is Fun

CONTINUITY

$f(x)$ is contiuous at $x = a$ if:

$$\lim_{x \to a} f(x) = f(a)$$

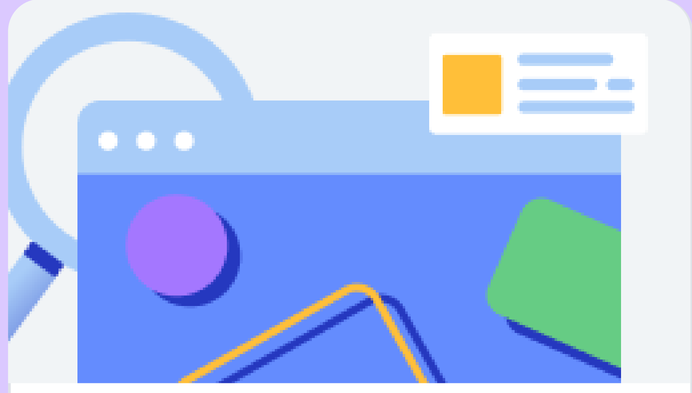
Is $f(x)$ continuous at $x = 0$?

$\boxed{a=0}$

$$f(x) = \begin{cases} (1+x)^{\frac{1}{x}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

$f(a)$ $f(0) = 0$

DIY



Limits to Infinity

Infinity is a very special idea. We know we cant reach it, but we can still try to work out the value of functions that have infinity

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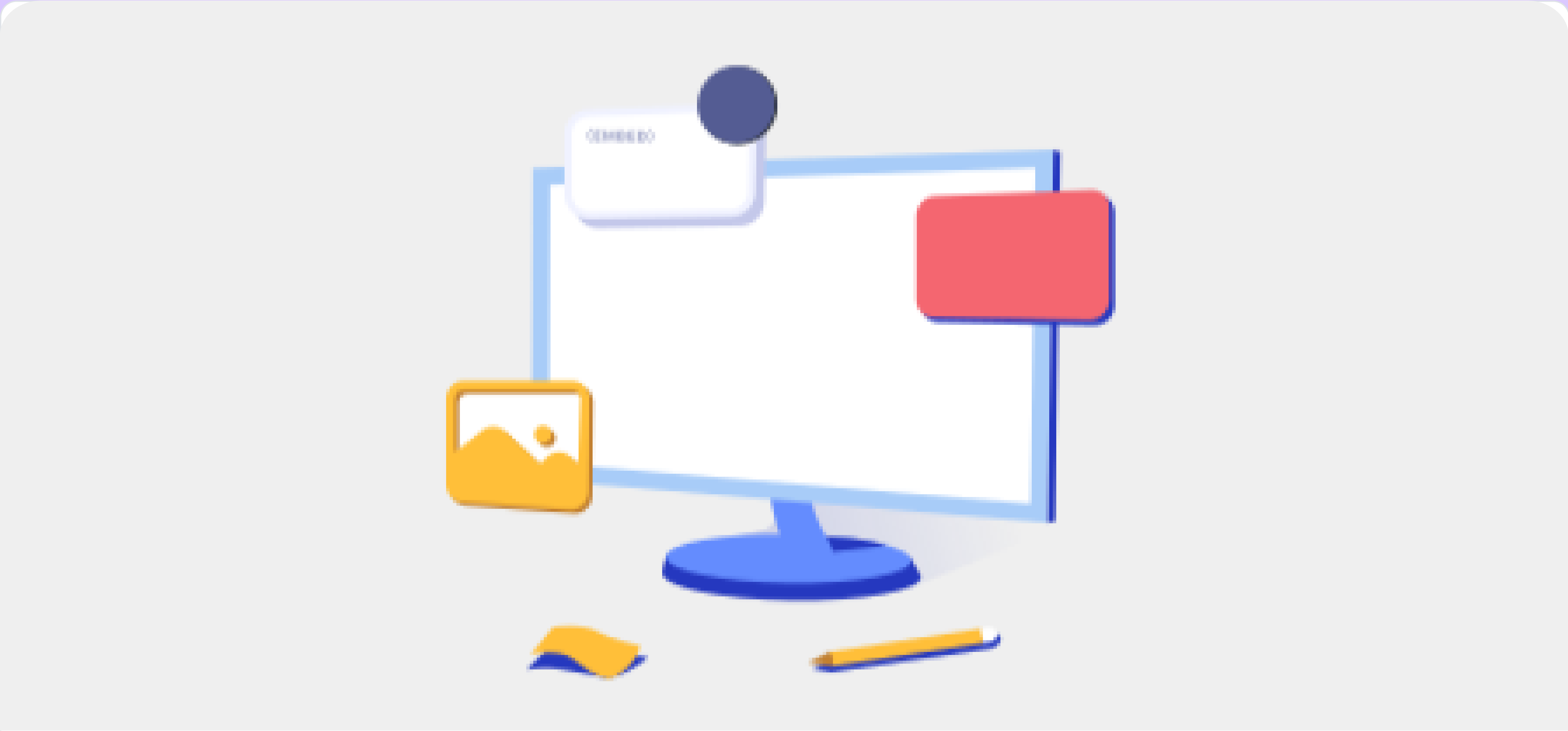
https://www.mathsisfun.com/calculus/limits-infinity.html



Limits to Infinity

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CONTINUITY

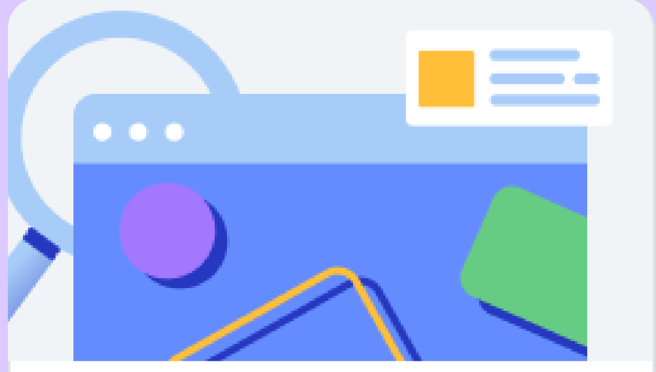
$f(x)$ is contiuous at $x = a$ if:

$$\lim_{x \to a} f(x) = f(a)$$

Is $f(x)$ continuous at $x = 0$?

$$f(x) = \begin{cases} x^2 sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

https://www.youtube.com/watch?v=7TGgWs_qCWY



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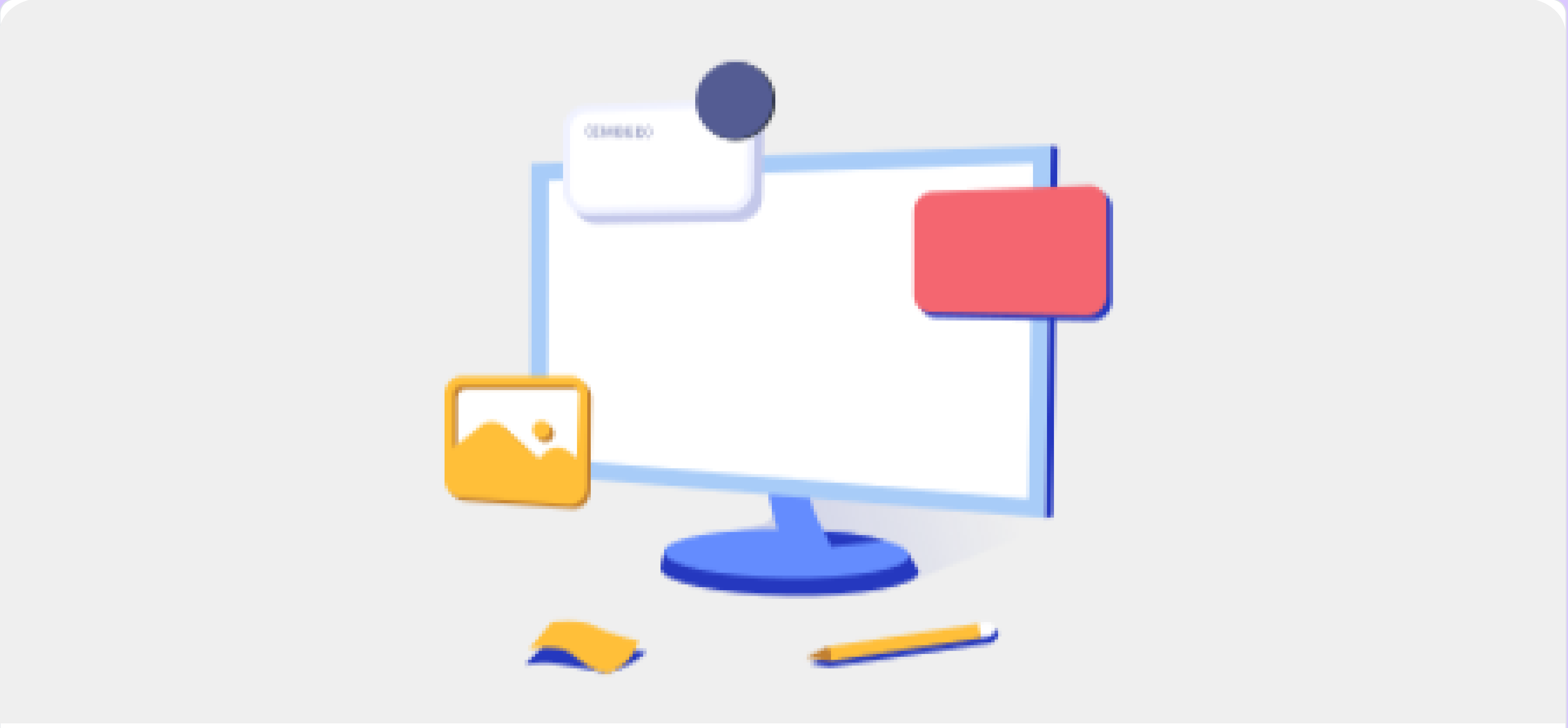

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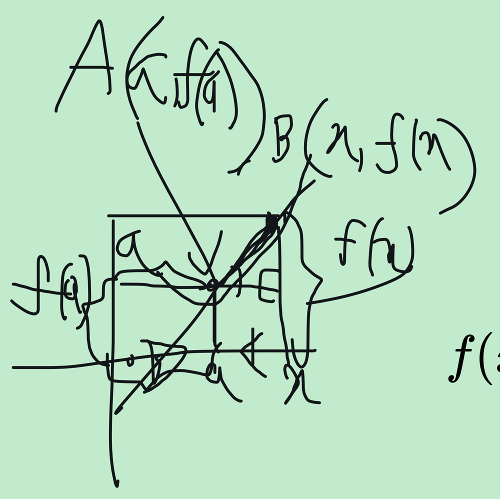


Slope $AB = \dfrac{f(x)-f(a)}{x-a} \Rightarrow B \to A \Rightarrow f'(a) = \lim\limits_{x\to a} \dfrac{f(x)-f(a)}{x-a}$ $\left[\dfrac{0}{0} \text{ form}\right]$

$x - a = h \Rightarrow x = a+h$

$x \to a \Rightarrow h \to 0$

$\Rightarrow f'(a) = \lim\limits_{h\to 0} \dfrac{f(a+h)-f(a)}{\cancel{a}+h-\cancel{a}}$

$\Rightarrow f'(a) = \lim\limits_{h\to 0} \left[\dfrac{f(a+h)-f(a)}{h}\right]$

$$\boxed{f'(x) = \lim_{h\to 0} \left[\frac{f(x+h)-f(x)}{h}\right]}$$

DIFFERENTIABILITY

$f(x)$ is differentiable at $x = a$ if:

$\lim\limits_{\substack{x\to a \\ (x\neq a)}} \dfrac{f(x)-f(a)}{x-a}$ *exists.*

Is $f(x)$ differentiable at $x = 0$?

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

DIY?

$\boxed{f(0) = 0}$

$f'(a) = \lim\limits_{x\to a} \left[\dfrac{f(x)-f(a)}{x-a}\right]$

$f'(0) = \lim\limits_{x\to 0} \left[\dfrac{x^2\sin(\frac{1}{x}) - f(0)}{x-0}\right]$

$\forall x \neq 0,$ $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$

$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$

$-\infty < \tan\left(\frac{1}{x}\right) < \infty$

$\forall x \neq 0, x \in \mathbb{R}$ $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$

$x^2(-1) \leq x^2\sin\left(\frac{1}{x}\right) \leq x^2 \cdot 1$ $[\because \forall x \in \mathbb{R}-\{0\} \Rightarrow x^2 > 0]$

$\boxed{\forall a,b,c \in \mathbb{R}-\{0\} \Rightarrow a<b \Rightarrow ac<bc}$

TRUE/FALSE

$2 < 3$

$2 \cdot \cancel{0} < 3 \cdot \cancel{0}$

$2 < 3$

$2c < 3c$

Let, $c = 4 \Rightarrow 2\cdot 4 = 8$ $3 \cdot 4 = 12$

$8 < 12$

$c = -4 \Rightarrow 2\cdot(-4) = -8$

$3\cdot(-4) = -12$

$2 < 3$

$\Rightarrow -8 > -12$

$-12 \ -8 \quad 0 \ 2 \ 3$

$\left[x^2\sin\left(\frac{1}{x}\right) \leftarrow x<0 \quad 0^- \to 0 \leftarrow 0^+ \quad x>0 \to x^2\sin\left(\frac{1}{x}\right)\right]$

$x \neq 0 \qquad x \to 0 \qquad x \neq 0$

$\Rightarrow f'(0) = \lim\limits_{x\to 0} \left[\dfrac{x^2\sin(\frac{1}{x}) - 0}{x-0}\right]$

$= \lim\limits_{x\to 0} \left[\dfrac{\cancel{x^2}\sin\left(\frac{1}{x}\right)}{\cancel{x}}\right]$

$f'(0) = \lim\limits_{x\to 0} \left[x\sin\left(\frac{1}{x}\right)\right]$ $\to \lim\limits_{x\to 0^+} \left[x\sin\left(\frac{1}{x}\right)\right] = 0$

$\to \boxed{\lim\limits_{x\to 0^-} \left[x\sin\left(\frac{1}{x}\right)\right]} = ?$ DIY

$\boxed{\forall x \in \mathbb{R}-\{0\}, \quad -1 \leq \sin\left(\frac{1}{x}\right) \leq 1}$

$x\cdot(-1) \leq x\sin\left(\frac{1}{x}\right) \leq x\cdot 1$ [Let, $\boxed{x > 0}$]

$\Rightarrow \lim\limits_{x\to 0^+} (-x) \leq \lim\limits_{x\to 0^+} \left[x\sin\left(\frac{1}{x}\right)\right] \leq \lim\limits_{x\to 0^+} (x)$

$[\because 0^+ > 0 \text{ and } 0^- < 0]$

$\Rightarrow -0 \leq \lim\limits_{x\to 0^+} \left[x\sin\left(\frac{1}{x}\right)\right] \leq 0$

$\Rightarrow 0 \leq \lim\limits_{x\to 0^+} \left[x\sin\left(\frac{1}{x}\right)\right] \leq 0$

$\Rightarrow$ By using Squeeze theorem,
(Sandwich)

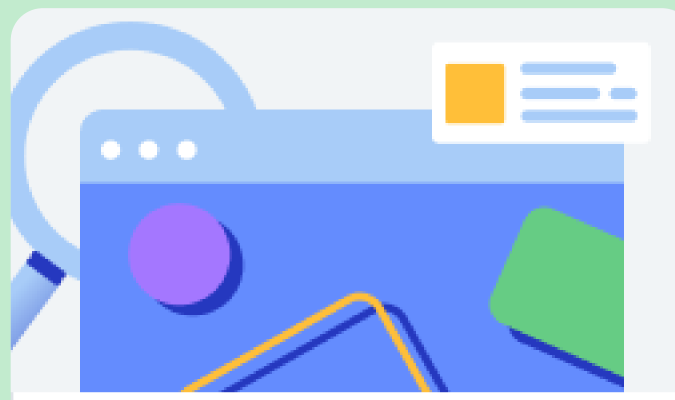
$$\boxed{\Rightarrow \lim_{x\to 0^+} \left[x\sin\left(\frac{1}{x}\right)\right] = 0} \Rightarrow \text{R.H.L}$$

L'HOPITAL'S RULE

If $f(x)$ and $g(x)$ is differentiable at either side of $x = a$, but not necessary at $x = a$ then:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

https://www.mathsisfun.com/calculus/l-hopitals-rule.html



L'Hopital's Rule

LHpitals Rule can help us calculate a limit that may otherwise be hard or impossible. ... LHpital is pronounced lopital. He was a French mathematician from the 1600s.

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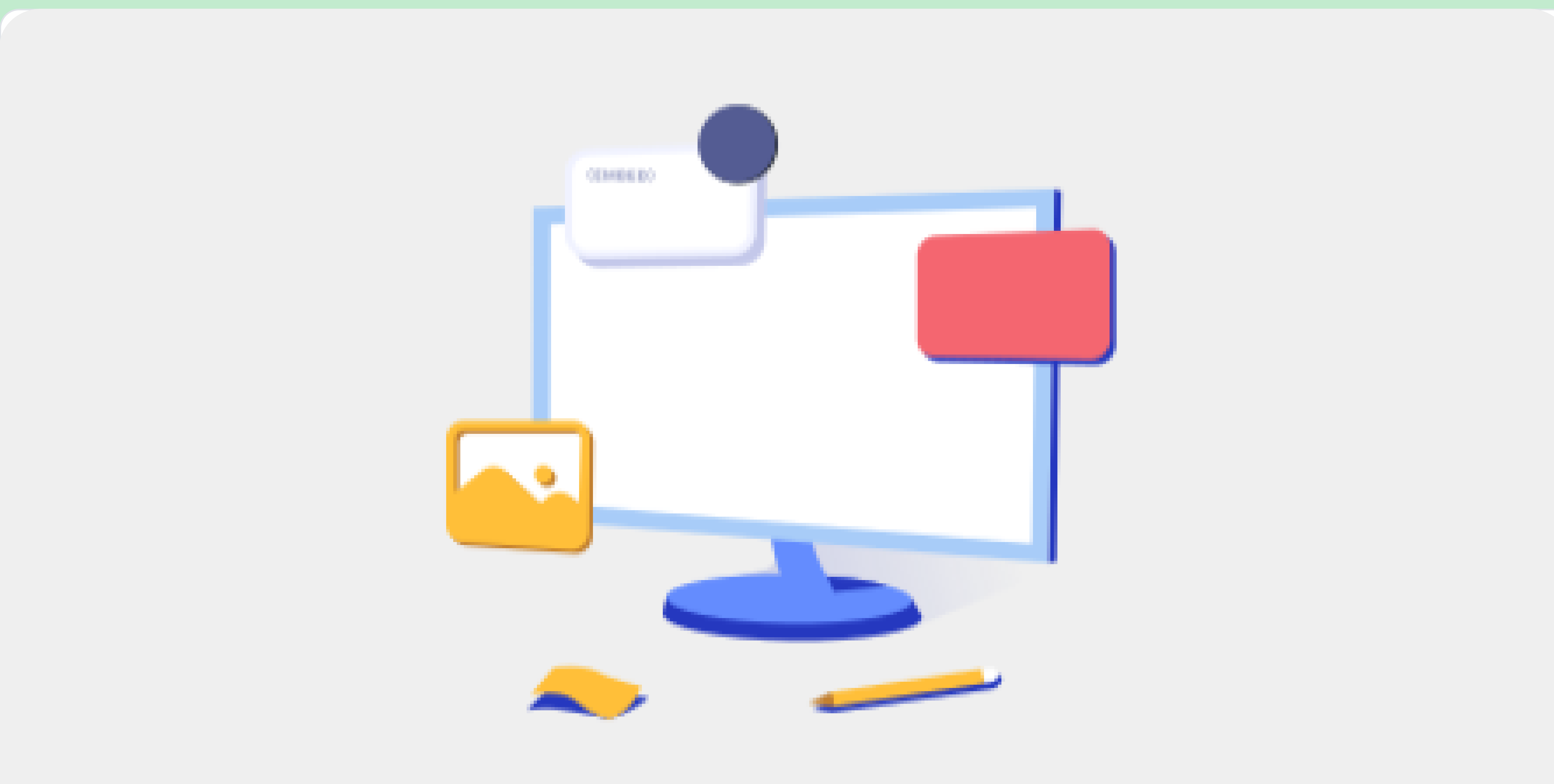

L'Hopital's Rule

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CHALLENGE: FIND THE MISTAKE HERE

$$\lim_{x \to 0} \frac{\sin x}{x}$$

$$= \lim_{x \to 0} \frac{\frac{d}{dx}(\sin x)}{\frac{d}{dx}(x)} \qquad \text{[Using L'Hô pital's Rule]}$$

$$= \lim_{x \to 0} \frac{\cos x}{1} \qquad \text{[Differentiating numerator and denominator]}$$

$$= \lim_{x \to 0} \cos x \qquad \text{[Simplifying the expression]}$$

$$= \cos(0) \qquad [\text{Substituting the value of } x = 0]$$

$$= 1 \qquad \text{[Final result]}$$